

Sentiment-Augmented Quarterly Stock Prediction

A leakage-aware supervised learning pipeline for predicting 63-day sector-relative stock performance after quarterly filing events

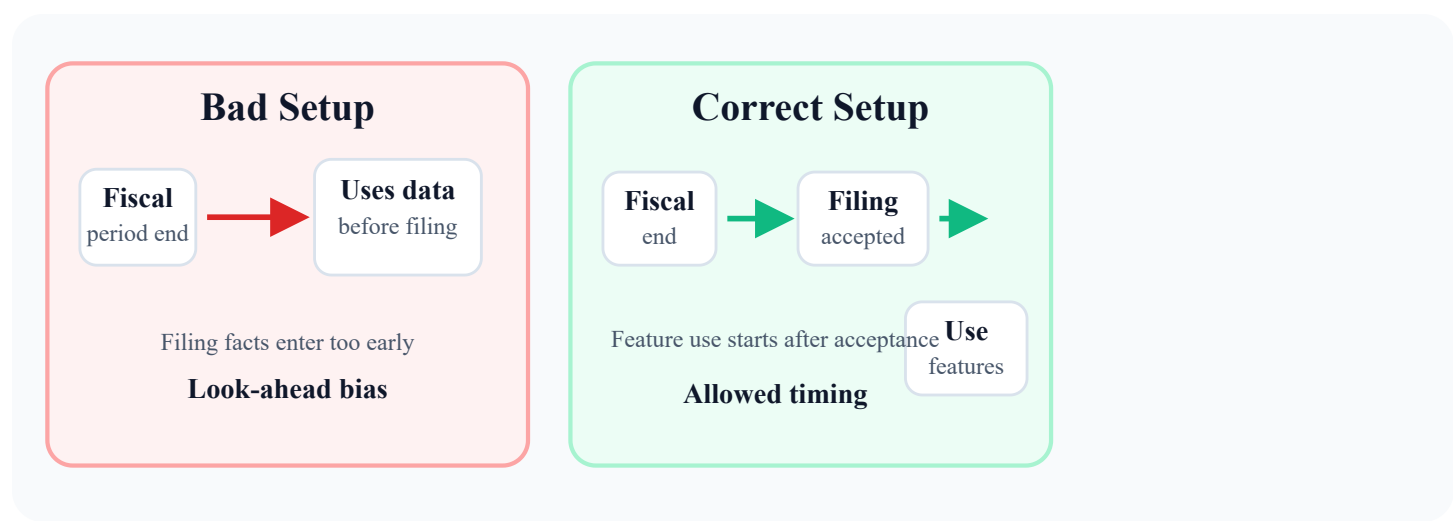
Thesis: This project tests whether accounting fundamentals, market context, and sentiment/event-specific features can predict whether a stock will outperform its sector after a quarterly filing event.

Why This Problem Is Difficult

Financial prediction is noisy and vulnerable to look-ahead bias. A model can appear predictive if it accidentally uses information that would not have been available at the time.

- No random train/test splits.
- No future information in features.
- Labels must not overlap validation windows.

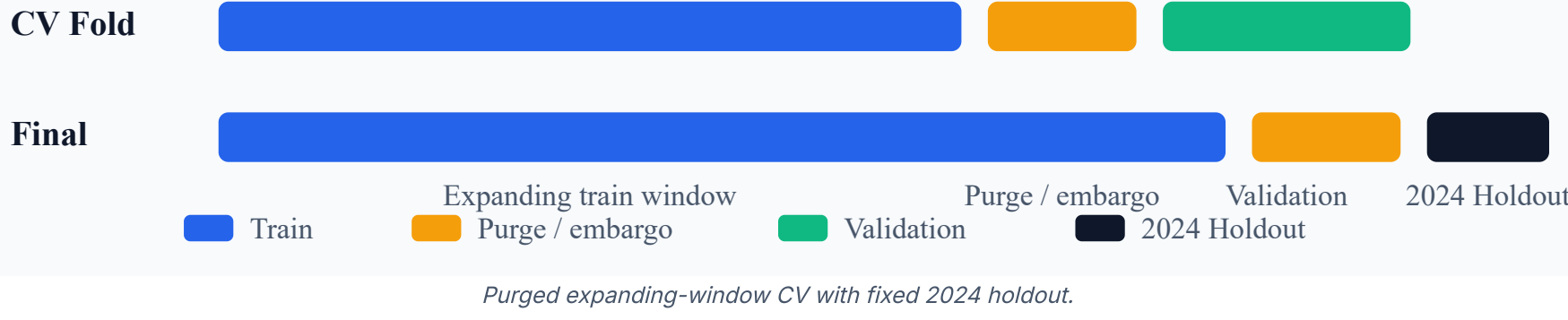
Leakage Control



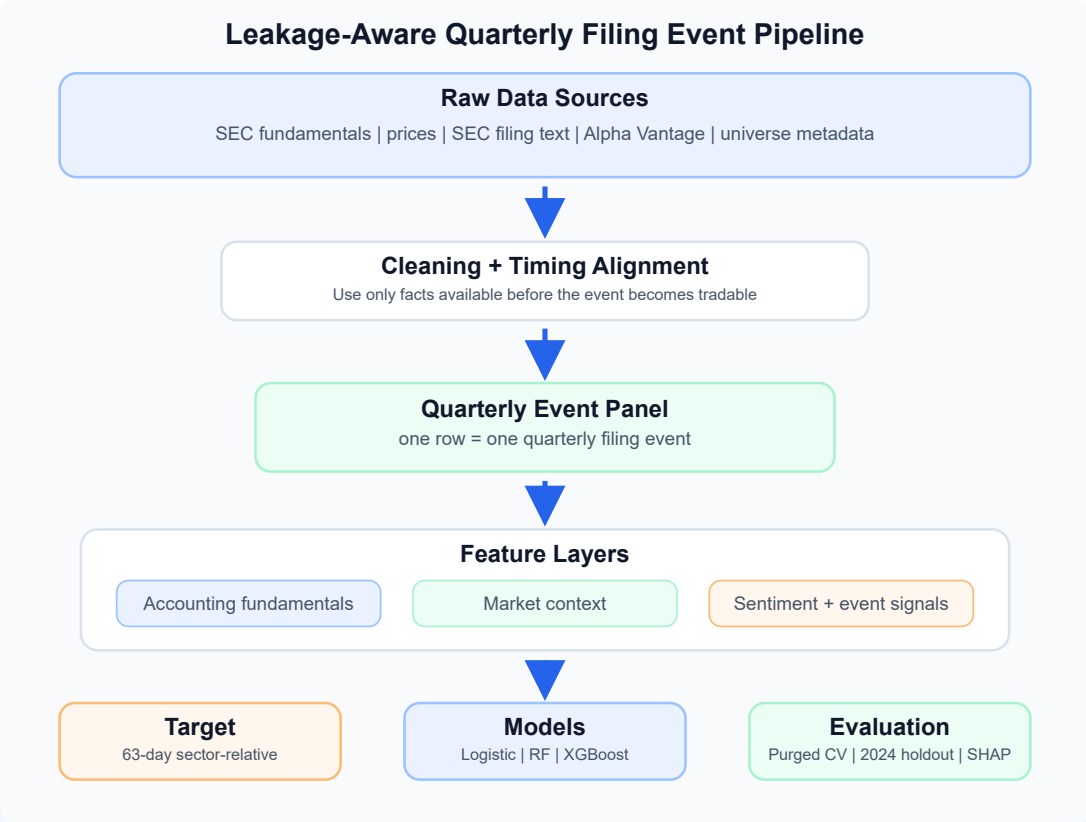
Event Timing & Validation



Purged Expanding-Window Validation

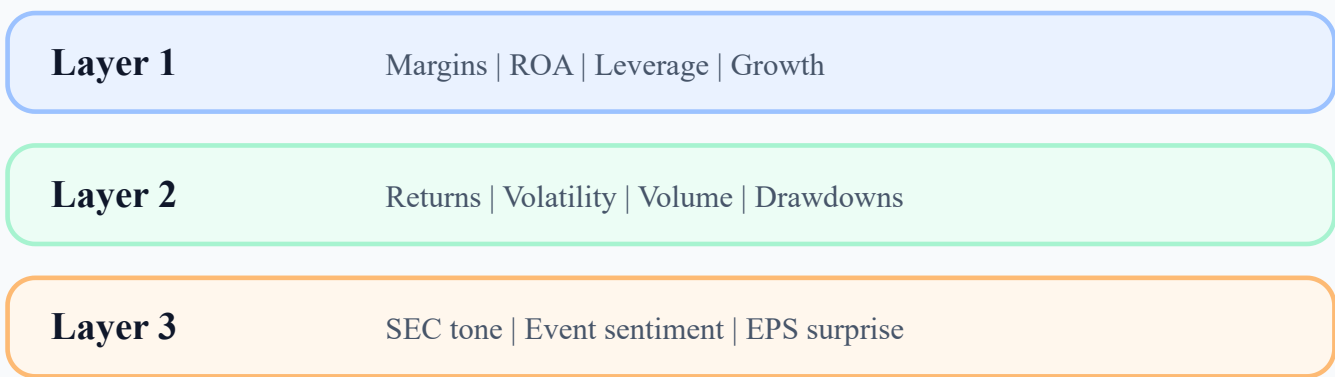


Modeling Pipeline



Feature Design

Controlled Feature-Addition Experiment



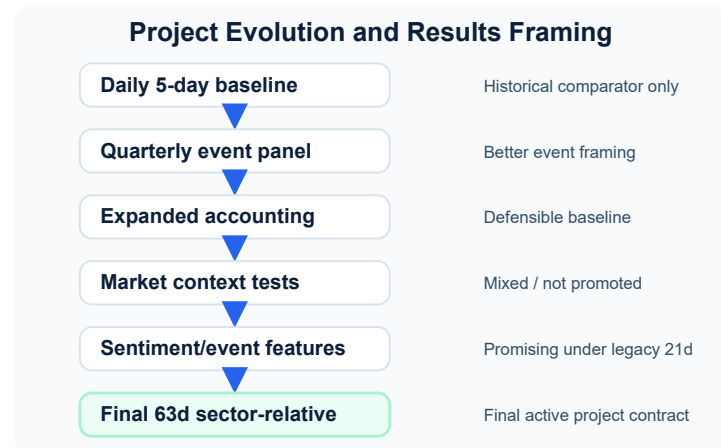
Target

1 if outperforming sector over 63 days, 0 otherwise

Algorithms Compared

- Logistic Regression (Baseline)
- Random Forest (Nonlinear)
- XGBoost (Boosted trees)

What the Experiments Showed

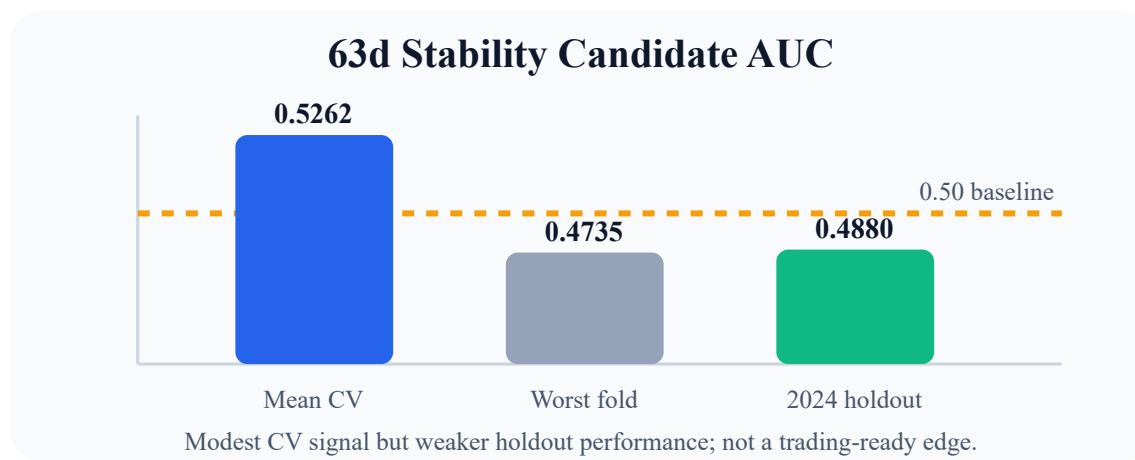


The strongest sentiment result comes from a legacy 21-day benchmark; the final contract uses the stricter 63-day sector-relative target.

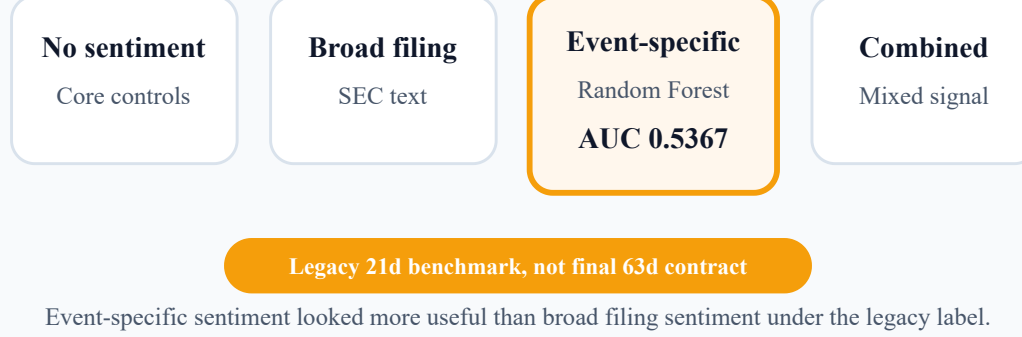
Selected Benchmark Evidence

63d Stability Candidate
XGBoost | CV AUC: 0.5262 | Holdout: 0.4880

Legacy 21d Sentiment
Random Forest | Holdout AUC: 0.5367
Legacy 21d benchmark, not final 63d contract.



Sentiment Feature Comparison



Main Findings

- Quarterly filing events are a better modeling unit than daily stock rows.
- Leakage control is central to the project.
- Fundamentals created a defensible baseline.
- Generic market features did not consistently improve results.
- Event-specific sentiment was the most promising incremental signal.

Conclusion & Next Steps

This is not a finished trading strategy. It is a leakage-aware research framework for testing whether financial signals add predictive value around quarterly filing events.

Next: Complete the final 63d rerun, improve data quality, expand event-specific sentiment, and defer portfolio construction until signal stability improves.

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